

Assignment 2

#Q1. How many Pillars of OOPs? Name them.

ANS:

* There are 4 pillars of OOPs namely:

1) Inheritance:-

- * Single / Simple Inheritance ✓
- * Multiple Inheritance ✓
- * Multilevel Inheritance ✓
- * Hierarchical Inheritance ✓
- * Hybrid Inheritance

2) Encapsulation:-

- * Private method ✓
- * Protected method ✓
- * Public method ✓

3) Data Abstraction ✓

4) Polymorphism ✓

- * Duck Typing ✓
- * Overriding ✓
- * Overloading ✓

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Qno.3) What is Encapsulation in OOPs?

ANS:-

Encapsulation means wrapping of data and methods into a single unit and restricting direct access to data.

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#Q21. Extract all vowels from the string using list comprehension: text "Comprehension"?

ANS:-

Method 1:

text = "Comprehension"

vowels = []

for i in text:

if i in "aeiouAEIOU":

vowels.append(i)

print(vowels)

Method 2: Using list comprehension.

text = "Comprehension"

[i for i in text if i in "aeiouAeiou"]

✓

2

✓

#Q20. Write a lambda that checks if a string starts with "A"? ["apple", "carrot", "Apple"]

ANS:-

```
b = ["apple", "carrot", "Apple"]  
start_with_A = lambda s: s.lower().startswith("a")  
result = list(filter(start_with_A, b))  
print(result)
```

#Q9. How many types of Methods in OOPs Name Them?

ANS:-

There are 3 types of methods in OOPs:

- 1) Instance method
- 2) Class method
- 3) Static method

#Q17. What is mode "w" and mode "a" in writing files?

The mode "w" in writing files means to write a data in a file.

The mode "a" in writing files means to append that is to add data in a file.

#Q16. Write a program to read CSV file using Python?

ANS:-

```
import csv  
file = open("C:\\Users\\DELL\\Desktop\\python.txt", "r")  
reader = csv.reader(file)  
print(reader)  
for row in reader:  
    print(row)
```

#Q15. Write code to checking files is exists or not in your pc drive?

ANS:-

try:

```
file = open("C:\\Users\\DELL\\Desktop\\python.txt", "r")  
content = file.read()  
print(content)  
file.close()
```

except FileNotFoundError:

```
    print("No such file")
```

else:

```
    print("Your code runned successfully")
```

#Q4. How many types of Inheritance? Name them.

ANS:-

There are 5 types of Inheritance in OOPs:

- 1) Single/Simple Inheritance:- A single child class inherits from single parent class.
- 2) Multiple Inheritance:- A child class inherits from more than one parent class.
- 3) Multilevel Inheritance :- This inheritance goes from level to level.
- 4) Hierarchical Inheritance :- In this type of inheritance many child class inherit from a single parent class.
- 5) Hybrid Inheritance :- It is a combination of two types of inheritance
example: Hierarchical inheritance + Multiple inheritance
Multiple Inheritance + Multilevel Inheritance

#Q8. How many types of Variables in OOPs?

ANS:-

There are 2 types of variable in OOPs:

- 1) Instance variable
- 2) Class variable

Qno.19 What is lambda function()? write names of lambda method.

ANS:-

A lambda function is a small anonymous (unnamed) function.

The names of lambda method are:

- 1) map()
- 2) filter()
- 3) reduce()
- 4) apply()

Qno.13) How many steps are there to read a file using python? write steps.

ANS:-

```
file = open("C:\\Users\\DELL\\Desktop\\python.txt", "r")  
content = file.read()  
print(content)  
file.close()
```

There are 4 steps to open a file :

- Step 1: To copy the path of a file.
- Step 2: Then open a file using open().
- Step 3: To read a file: read().
- Step 4: To close a file: close()

Qno.14) How many steps are there to write a file using python? write steps.

ANS:-

```
file_path = r"C:\\Users\\DELL\\Desktop\\python.txt"
```

```
file = open(file_path, "w")
```

```
file.write("\n Hello")
```

```
file.close()
```

There are 4 steps to write a file using python:

Step 1: To copy the path of a file.

Step 2: To open a file .

Step 3: To write a data in a file: write()

Step 4: To close a file: close()

Qno12) What is super() in OOPs?

ANS:-

The super() in OOPs is used to call the parent class inside the child class.

Qno.5) How many types of case ? Name them and give one example each

ANS:-

There are 3 types of case in python namely:

1) snake_case:

Example :

2) PascalCase:

Example:

3) camelCase:

Example:

Qno18) What is difference between "w" and "x" mode in writing files ?

ANS:-

The difference between the "w" and "x" mode in writing files in python is:

The "w" mode in writing files is used to create a file whereas the "x" mode in writing file is used to create a file if error exist.